

Q1. Match the following topics with their most suitable characteristics:

1. Probability
 2. Conic Sections: Ellipse
 3. Units and Measurements
- I. Involves calculating likelihoods based on prior knowledge and certain conditions
 - II. Focuses on accuracy and precision in science and technology through standardized measurement systems
 - III. Includes the study of shapes described by quadratic equations in standard form
- A. Bayes' theorem
 - B. SI fundamental and derived units
 - C. Equation of an ellipse in standard form
- A. 1-A, 2-C, 3-B
 - B. 1-C, 2-A, 3-B
 - C. 1-B, 2-C, 3-A
 - D. 1-A, 2-B, 3-C

Answer: Option 1

Solution: Probability is closely linked with Bayes' theorem, which uses prior knowledge to update probabilities. The equation of an ellipse is a fundamental aspect of studying ellipses under conic sections. Units and Measurements heavily involve understanding and using SI units to ensure precision. Hence, Option (1) is the correct answer.

Q2. Match the following topics with their corresponding analytical approaches:

1. Probability
 2. Conic Sections: Ellipse
 3. Units and Measurements
- I. Requires understanding of theoretical and empirical relationships between physical quantities
 - II. Involves geometric and algebraic manipulation to derive properties
 - III. Includes computations based on conditional probabilities and statistical inference
- A. Dimensional analysis and applications
 - B. Probability distribution of a random variable
 - C. Basic coordinate-geometric treatment
- A. 1-B, 2-C, 3-A
 - B. 1-A, 2-B, 3-C
 - C. 1-C, 2-A, 3-B
 - D. 1-B, 2-A, 3-C

Answer: Option 1

Solution: Probability often deals with statistical inference and the distribution of random variables. The ellipse is explored through its geometric and algebraic properties. Units and Measurements require a thorough understanding of dimensional analysis to relate different physical quantities. Hence, Option (1) is the correct answer.

Q3. Match the following topics with suitable mathematical or scientific methods:

1. Probability
 2. Conic Sections: Ellipse
 3. Units and Measurements
- I. Employs statistical methods to analyze the likelihood of outcomes
 - II. Utilizes coordinate geometry for shape analysis
 - III. Involves the systematic study of measurement systems for accuracy
- A. Measurement errors and significant figures

- B. Addition and multiplication theorems
- C. Equation of an ellipse in standard form
- A. 1-B, 2-C, 3-A
- B. 1-A, 2-B, 3-C
- C. 1-C, 2-A, 3-B
- D. 1-B, 2-A, 3-C

Answer: Option 1

Solution: Probability uses addition and multiplication theorems to determine the probability of events. The study of ellipses involves detailed understanding of their equations in standard form, which is a crucial aspect of coordinate geometry. Units and Measurements deal with ensuring the correctness of measurements through understanding errors and significance of figures. Hence, Option (1) is the correct answer.